TUGAS MOBILE WEEK 7 GESTURE APPS

MOCH. RAMZY NAUFAL ENDI PUTRA

main.dart

gesture > lib > main.dart > ...

```dart
import 'package:flutter/material.dart';
import 'home_page.dart';

Run | Debug | Profile
void main() {
  runApp(GestureApp());
}

class GestureApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Gesture Lab',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(useMaterial3: true, colorSchemeSeed: Colors.blue),
      home: HomePage(),
    ); // MaterialApp
  }
}
```

main.dart

File yang berfungsi mendefinisikan awal ketika flutter di run

home_page.dart

File yang dipanggil file main.dart, file ini berfungsi mengatur isi content

home_page.dart

gesture > lib > home_page.dart > ...

```dart
import 'package:flutter/material.dart';
import 'pages/tap_page.dart';
import 'pages/drag_page.dart';
import 'pages/scale_page.dart';
import 'pages/playground_page.dart';

class HomePage extends StatelessWidget {
  final menus = [
    {"title": "Tap Interaction", "page": TapPage()},
    {"title": "Drag Movement", "page": DragPage()},
    {"title": "Scale & Zoom", "page": ScalePage()},
    {"title": "Gesture Playground", "page": PlaygroundPage()},
  ];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("Gesture Lab")),
      body: Padding(
        padding: EdgeInsets.all(16),
        child: GridView.builder(
          itemCount: menus.length,
          gridDelegate: SliverGridDelegateWithFixedCrossAxisCount(
            crossAxisCount: 2,
            crossAxisSpacing: 16,
            mainAxisSpacing: 16,
          ), // SliverGridDelegateWithFixedCrossAxisCount
          itemBuilder: (_, i) {
            return GestureDetector(
              onTap: () {
                Navigator.push(
                  context,
                  MaterialPageRoute(builder: (_) => menus[i]["page"] as Widget),
                );
              },
              child: AnimatedContainer(
                duration: Duration(milliseconds: 300),
```

tap_page.dart

gesture > lib > pages > tap_page.dart > ...

```dart
import 'package:flutter/material.dart';

class TapPage extends StatefulWidget {
  @override
  _TapPageState createState() => _TapPageState();
}

class _TapPageState extends State<TapPage> {
  double size = 150;
  Color color = Colors.blue;

  void randomize() {
    setState(() {
      size = size == 150 ? 220 : 150;
      color = color == Colors.blue ? Colors.red : Colors.blue;
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("Tap Interaction")),
      body: Center(
        child: GestureDetector(
          onTap: randomize,
          onDoubleTap: () => setState(() => color = Colors.green),
          onLongPress: () => setState(() => size = 250),
          child: AnimatedContainer(
            duration: Duration(milliseconds: 300),
            width: size,
            height: size,
            decoration: BoxDecoration(
              color: color,
              borderRadius: BorderRadius.circular(30),
              boxShadow: [BoxShadow(blurRadius: 20, color: Colors.black26)],
            ), // BoxDecoration
          ), // AnimatedContainer
```

tap_page.dart

File yang dipanggil di file home_page sebelumnya file ini bertugas untuk mengatur logika tentang gesture tap

drag_page.dart

File yang dipanggil pada file home_page untuk mengatur logika tentang gesture drag

home_page.dart

gesture > lib > home_page.dart > ...

```dart
import 'package:flutter/material.dart';
import 'pages/tap_page.dart';
import 'pages/drag_page.dart';
import 'pages/scale_page.dart';
import 'pages/playground_page.dart';

class HomePage extends StatelessWidget {
  final menus = [
    {"title": "Tap Interaction", "page": TapPage()},
    {"title": "Drag Movement", "page": DragPage()},
    {"title": "Scale & Zoom", "page": ScalePage()},
    {"title": "Gesture Playground", "page": PlaygroundPage()},
  ];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("Gesture Lab")),
      body: Padding(
        padding: EdgeInsets.all(16),
        child: GridView.builder(
          itemCount: menus.length,
          gridDelegate: SliverGridDelegateWithFixedCrossAxisCount(
            crossAxisCount: 2,
            crossAxisSpacing: 16,
            mainAxisSpacing: 16,
          ), // SliverGridDelegateWithFixedCrossAxisCount
          itemBuilder: (_, i) {
            return GestureDetector(
              onTap: () {
                Navigator.push(
                  context,
                  MaterialPageRoute(builder: (_) => menus[i]["page"] as Widget),
                );
              },
              child: AnimatedContainer(
                duration: Duration(milliseconds: 300),
```

scale_page.dart

gesture > lib > pages > scale_page.dart > ...

```dart
import 'package:flutter/material.dart';
import 'package:flutter/gestures.dart';

class ScalePage extends StatefulWidget {
  const ScalePage({super.key});

  @override
  State<ScalePage> createState() => _ScalePageState();
}

class _ScalePageState extends State<ScalePage> {
  double scale = 1.0;
  double previousScale = 1.0;

  Offset position = Offset.zero;

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text("Scale & Zoom (PRO)")),
      body: Center(
        child: Listener(
          onPointerSignal: (pointerSignal) {
            // ✅ Zoom pakai scroll (laptop / mouse)
            if (pointerSignal is PointerScrollEvent) {
              setState(() {
                scale += pointerSignal.scrollDelta.dy * -0.001;
                scale = scale.clamp(0.5, 3.0);
              });
            }
          },
          child: GestureDetector(
            // ✅ Start pinch
            onScaleStart: (details) {
              previousScale = scale;
            },
```

scale_page.dart

File yang dipanggil pada file home, yang bertugas untuk mengatur logika tentang gesture scale

playground_page.dart

File yang dipanggil di home_page, yang bertugas untuk mengatur logika tentang gesture playground

home_page.dart

gesture > lib > home_page.dart > ...

```dart
import 'package:flutter/material.dart';
import 'pages/tap_page.dart';
import 'pages/drag_page.dart';
import 'pages/scale_page.dart';
import 'pages/playground_page.dart';

class HomePage extends StatelessWidget {
  final menus = [
    {"title": "Tap Interaction", "page": TapPage()},
    {"title": "Drag Movement", "page": DragPage()},
    {"title": "Scale & Zoom", "page": ScalePage()},
    {"title": "Gesture Playground", "page": PlaygroundPage()},
  ];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("Gesture Lab")),
      body: Padding(
        padding: EdgeInsets.all(16),
        child: GridView.builder(
          itemCount: menus.length,
          gridDelegate: SliverGridDelegateWithFixedCrossAxisCount(
            crossAxisCount: 2,
            crossAxisSpacing: 16,
            mainAxisSpacing: 16,
          ), // SliverGridDelegateWithFixedCrossAxisCount
          itemBuilder: (_, i) {
            return GestureDetector(
              onTap: () {
                Navigator.push(
                  context,
                  MaterialPageRoute(builder: (_) => menus[i]["page"] as Widget),
                );
              },
              child: AnimatedContainer(
                duration: Duration(milliseconds: 300),
```